Understanding Sociocultural Alignment in Multilingual LLMs via Natural Debate Statements

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Abstract

In this work we investigate the sociocultural values learned by large language models (LLMs). We introduce *CultureProbe*, a novel open-access dataset constructed from natural debate statements using a multi-step methodology. The dataset is synthetically annotated with quantized sociocultural norms according to the Hofstede cultural dimensions. We verify the accuracy of the synthetic annotations using a crowdsourced human annotation experiment conducted on a representative sample. We conduct a comparative analysis between two groups of LLMs developed in different countries (U.S. and China), and use as a comparative baseline patterns observed in human measurements. Using this new dataset and the analysis above, we found that *culturally-distinct LLMs largely reflect the values and norms of the countries in which they were developed*, highlighting their inability to adapt to the sociocultural backgrounds of their users.

Note: Our focus is on the identification of sociocultural biases in LLMs. It is not our intention to compare cultures.

1 Introduction

Over the past few years, rapid advancements have pushed the performance boundaries of large language models (LLMs), driven by large-scale pre-training and post-training alignment methods, such as instruction tuning and reinforcement learning from human feedback. These models are widely used by hundreds of millions of people for a variety of tasks (Chatterji et al., 2025).

However, as noted in prior work (Johnson et al., 2022; Shen et al., 2024; Mihalcea et al., 2025; Pawar et al., 2025), general-purpose AI systems often fail to adapt to the diverse sociocultural backgrounds and value systems of their users. This creates the risk of imposing a set of ostensibly universal beliefs, potentially shaping and distorting the sociocultural values and norms of their users (Jakesch et al., 2023; Chawla et al., 2023). Despite existing research advocating for multicultural annotation practices (Plank, 2022; Prabhakaran et al., 2022), the development of LLMs continues to rely predominantly on a single alignment paradigm.

Here, we quantify the extent to which *LLM responses reflect the sociocultural norms of the countries in which they were developed*. Our analysis focuses on LLMs originating from two of the most influential countries in current LLM development: the United States and China. In particular, the contributions of our paper are:

(1) We introduce *CultureProbe*, a dataset obtained from naturally-occurring debates. These debate statements were created by a wide set of international users of the Kialo debate platform[1] on universally-meaningful topics. We synthetically annotate this dataset with quantized sociocultural values grounded in Hofstede's cultural dimensions (Hofstede, 1991). We verify the quality of these annotations using human quality control on a representative sample. To our knowledge, *CultureProbe* is substantially larger than prior datasets on Hofstede sociocultural norms, containing 985 distinct debate statements, which expand into 2007 dimension annotations. The dataset is publicly available (see Appendix A).

(2) Using *CultureProbe*, we assess the sociocultural values expressed by multiple LLMs developed in two countries (the U.S. and China) under a multilingual prompting setup (English and Chinese). We perform comparative analyses across multiple levels of granularity: (a) a comparison of aggregated responses at the sociocultural group level; (b) an examination of sociocultural biases at the individual model level; and (c) an analysis of how the prompting language influences the sociocultural values expressed by each model.

[1]https://www.kialo.com

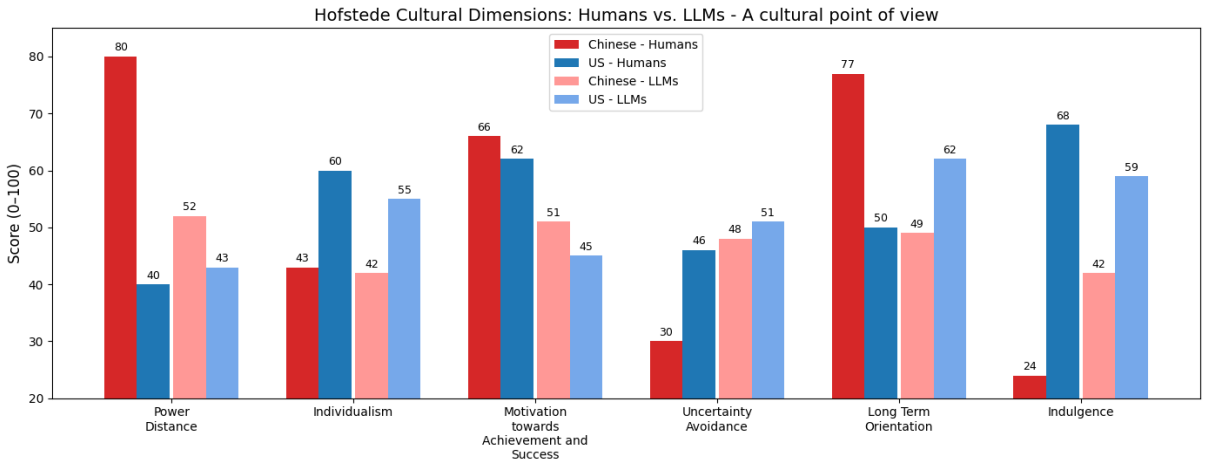


Figure 1: Scoring of humans and LLMs according to the Hofstede dimensions. LLMs were prompted to agree or disagree with a given statement, and each response was assigned a polarity score ranging from -2 to $+2$ along the corresponding Hofstede dimension. These scores were then averaged and linearly mapped to a 0–100 scale for readability. We observe that the LLMs largely follow the trends observed in human measurements across five out of six dimensions. This figure is a summary of our complete results. Please see Appendix E for all results.

(3) We compare our results against a baseline derived from human measurements, and highlight important observations, grounded in social science theory. For example, one of our main observations, as illustrated in Figure 1, is that culturally-distinct LLM groups follow trends similar to those observed among people from the same culture, across five of the six Hofstede dimensions. Moreover, we observed no statistically significant differences in the sociocultural values elicited when prompting the same LLM in English and Chinese.

2 Related work

Motivated by the view that language is fundamentally embedded in social and cultural context (Yang et al., 2025; Shen et al., 2024), recent works try to examine whether LLMs encode sociocultural values as latent properties of model behavior. Analyses of free-form generations in culturally salient scenarios indicate that although LLMs exhibit culturally patterned tendencies, these signals appear to be weak, context-dependent, and internally inconsistent, often defaulting to dominant norms rather than reflecting stable cultural value representations.

Several works ground cultural evaluation in established social science frameworks, most notably Hofstede's cultural dimensions (Hofstede, 1991). Kharchenko et al. (2025) introduces a large-scale multilingual advice benchmark covering five Hofstede dimensions and 36 country–language pairs, combining manually constructed (single annotator) and LLM-generated prompts, persona conditioning, and multilingual translation. Although LLMs distinguish opposing sociocultural values, they fail to apply them consistently in advice, even under explicit country personas, exposing a gap between conceptual recognition and behavioral expression. *CultureProbe* is built from naturally occurring debate statements and supported by a substantially stronger human validation procedure. It covers all six Hofstede dimensions and contains 2,007 dimension-aligned statements per language, compared to 250 prompts per language in prior work. Moreover, assessing cultural alignment through explicit nationality-based persona prompting may conflate the elicitation of cultural knowledge with underlying sociocultural representations. By contrast, we probe sociocultural values implicitly by analyzing how LLMs evaluate the truthfulness of natural statements aligned with Hofstede's dimensions. This indirect evaluation framework enables the study of sociocultural value representations in settings that more closely resemble real-world LLM interactions, avoiding potential biases introduced by explicit cultural conditioning.

Using a standardized survey-based approach, Masoud et al. (2025) operationalize the Hofstede VSM13 questionnaire to measure cultural alignment in LLMs. By varying persona prompts, decoding parameters, and language-specific fine-tuning, they show that cultural alignment is weak and unstable across models; while GPT-4 performs best, results are highly sensitive to prompt and sampling choices, indicating fragile rather than robust cultural representations. In contrast, our work investigates whether sociocultural values emerge im-

plicitly through model evaluations of natural statements, rather than via explicit persona-based and survey-driven cultural elicitation, while also relying on a substantially larger dataset of naturally occurring, human-validated statements.

Finally, Ma et al. (2025) argue that closed-ended evaluations underestimate LLMs' social reasoning. Drawing on survey methodology and qualitative social science, they advocate for open-ended generation to better capture interpretability and heterogeneity in social simulation, while highlighting open challenges in benchmarking, evaluation reliability, and ethical risks posed by plausible but unverifiable outputs. Our work operationalizes several principles advocated in this line of research by reducing explicit cultural conditioning and improving ecological validity through the use of naturally occurring discourse instead of rigid survey reconstruction and persona-based prompting.

Complementing these research efforts, practitioners have also conducted informal experiments to assess AI's cultural competence: the Culture Factor Group[2] conducts an exploratory cultural audit by manually designing a 42-question instrument aligned with Hofstede's 6 dimensions and prompting several LLMs (e.g. GPT-4, Gemini) with explicit national personas, with and without theory priming, revealing substantial inconsistencies between expected and model-generated cultural profiles (The Culture Factor, 2024). Our framework introduces a substantially larger and more systematically validated dataset based on naturally occurring discourse, enabling implicit and ecologically grounded probing of sociocultural values, without relying on handcrafted persona-based prompting.

3 The Hofstede cultural dimensions

Our analysis relies on the Hofstede cultural dimensions framework (Hofstede, 1991). For completeness, we provide brief definitions of its dimensions in Table 1, also specifying the traits of each polarity (i.e., negative and positive).

4 Dataset construction and alignment

At a high-level, *CultureProbe*, our proposed dataset, aligns natural debate statements with the six Hofstede cultural dimensions. We collected debate statements from the Kialo online debate platform.[1] These debate statements were created by a wide set of international Kialo users on universally-meaningful topics; many statements are largely relevant for one or more Hofstede dimensions. For example, supporting the statement *"Countries should invest in renewable energies"* indicates strong alignment with the Long Term Orientation (LTO) dimension. We synthetically annotate these statements through a multi-step annotation pipeline. Our annotations provide scores that can measure the alignment of agreeing/disagreeing with a statement to a certain Hofstede dimension. Figure 2 describes the dataset generation process, further detailed below.

4.1 Statement extraction, filtering, and translation

Kialo is a web platform designed for structured debate and argument analysis, consisting of thousands of individual debates on a broad area of subjects: politics, philosophy, science, ethics, religion. Each debate starts with a general statement (e.g., *"Wealthy countries should provide citizens with a universal basic income"*); users may provide arguments in favor or against that statement. A voting system ensures statement relevance and the strength of the arguments in the debate process.

We construct our dataset by first scraping these debates, from which we extract the initial statements and the number of votes. This vote count is then used as a filtering measure, to ensure the quality and relevance of the statement in our dataset. That is, since statements with more votes reflect a higher degree of user approval on their perceived reliability, we ensure that we filter out content that is potentially harmful or irrelevant. In particular, in this work, we use debates that have at least one vote. We include examples of the scraped statements in Appendix A. We manually inspected a sample of the downloaded statements and found no instances of spam or other form of abuse.

We translated all statements into Chinese using the googletrans library. A sample was verified by a native Chinese speaker, who indicated that all translations were correct.

4.2 Statement annotation

The selected statements undergo a multi-step synthetic annotation process. In each step we aggregate the responses of several culturally-different LLMs (i.e., U.S. and Chinese), also prompting each in two different languages: English and Chinese. We used GPT, Llama, DeepSeek and Qwen

[2] https://www.theculturefactor.com/

Dimension	Definition	Positive polarity	Negative polarity
Power Distance Index (PDI)	Evaluates the degree to which citizens tolerate unequal power relationships between various groups.	acceptance of power inequality	equalitarian values
Collectivism vs. Individualism (IDV)	Evaluates the degree to which a country organizes people into homogeneous groups that have specific duties and obligations within the social structure.	individualistic pursuit	greater good of groups
Uncertainty Avoidance (UAI)	Evaluates the extent to which the members of a culture feel threatened by ambiguous or unknown situations.	high levels of risk aversion	risks acceptance
Motivation towards Achievement and Success (MAS)	Formerly Masculinity vs. Femininity, it evaluates societal tendencies regarding gender roles, gender equality, and traditional gender values.	goal-oriented behavior, wealth accumulation, clearly defined gender roles.	collective quality of life, and flexible gender roles
Short-term vs. Long-term Orientation (LTO)	Evaluates the degree to which a society acts to further short-term objectives at the expense of long-term goals and vice versa.	immediate gratification	sacrifice of instant results for the greater long-term good
Indulgence vs. Restraint (IND)	Evaluates the degree to which a society allows free gratification of basic and natural human desires related to enjoying life, as opposed to restraining through strict social norms and regulations.	life enjoyment, entertainment, luxuries, and showy displays of socioeconomic status.	importance of modesty, suppression, and self-control.

Table 1: Hofstede dimensions definitions together with interpretations of response polarities.

as synthetic annotators. The models' identifiers are presented in subsection 6.1.

This "LLM-as-a-judge" process is conducted in parallel with a crowdsourced experiment on a representative sample of statements, involving multicultural human annotators. We use this multicultural, multilingual evaluation approach in order to account for cultural and linguistic variability, which allows us to obtain a cross-cultural collection of human annotations for the same labeling tasks performed by the LLMs. These annotations are then used to assess the quality and reliability of the synthetic labels. See subsection 4.3 for details.

These "wisdom-of-the-crowd" annotations are designed to capture the central tendency for each sociocultural dimension that applies to a given statement. This allows us to inspect how polarized individual models are (second part of this paper).[3]

Our two-step labeling pipeline is guided by the following questions:

(1) Is a given statement relevant to a particular Hofstede dimension?

Scraped statements have a wide variety with respect to the societal aspects they address. Moreover, as the Hofstede dimensions are not mutually exclusive, some statements may be relevant for more than one dimension, while other may not meaningfully align with any of them. For instance, agreement with the statement *"Governments should make an effort to reduce the gender pay gap"* is informative for PDI and MAS, while being unrelated to the IND and UAI dimensions.

To robustly annotate this non-trivial one-to-many alignment between statements and Hofstede dimensions, we decompose the annotation process into a series of binary questions, i.e., is statement X relevant for dimension Y? The four LLMs previously mentioned are prompted in both English and Chinese to answer the above question for each combination of statement and dimension in our dataset. We then aggregate the responses and keep only pairs of statement/dimension that were labeled as relevant in at least three of the eight predictions.

(2) Does agreeing with a statement indicate positive or negative polarity for a certain Hofstede dimension?

Next, we quantify where agreement with a specific claim falls on a discrete polarity scale for a given cultural dimension. For example, agreeing with *"Billionaires should be taxed at 100% on all wealth over $999,999,999"* strongly aligns with the negative polarity of the IDV dimension.[4]

We employ a discrete scoring system with the following possible values and interpretations (for agreement with the statement):

-2**:** strong negative polarity

-1**:** slight negative polarity

$+1$**:** slight positive polarity

$+2$**:** strong positive polarity

Since statements are filtered for relevance in the previous step, a score of 0 (neutral) is not permitted.

Scores produced by the multicultural and multilingual set of LLMs are aggregated into a single statement-level score per relevant dimension by taking the median of the score distribution. We chose

[3] Understanding the variation of individual human annotators is another very interesting research direction, which we leave for future work.

[4] In this work, we consider disagreement with a statement to have the opposite score as agreement. Initial experiments with human annotators supported this assumption.

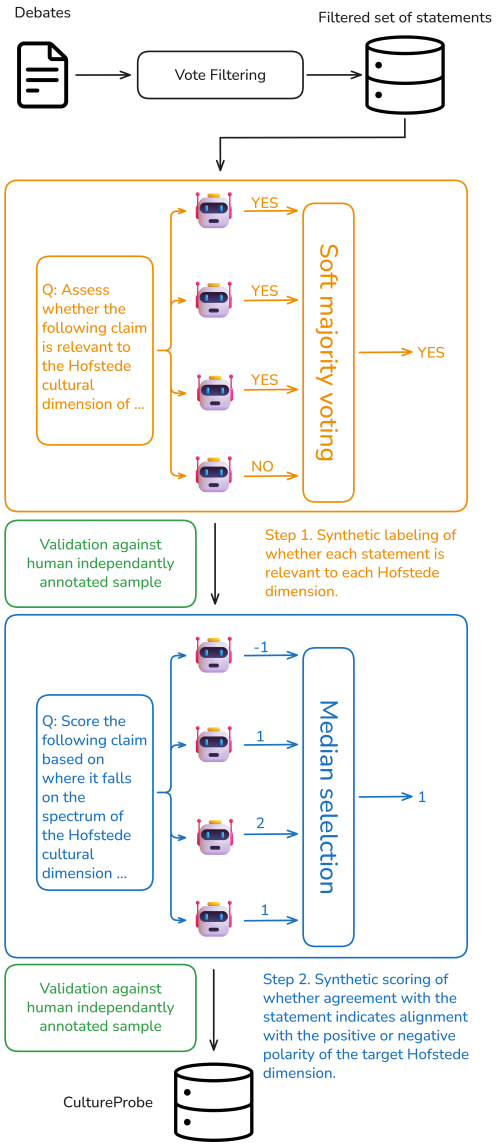


Figure 2: The statement scoring pipeline based on the six Hofstede dimensions. We collect natural debate statements via web scraping and filter them based on vote counts. Step 1 further removes statements that are irrelevant to the Hofstede dimensions. Step 2 assigns a score (-2 to $+2$), indicating how agreement with a statement aligns with the negative or positive polarity of each Hofstede dimension. To validate this process, we compared the synthetic labels against human annotations collected through a parallel crowdsourced experiment.

the median rather than the mean as it is less sensitive to outliers, and it reflects the central tendency of scores across culturally situated model outputs.

4.3 Validation against human annotations

To understand the quality of the synthetic annotations, we repeated the two annotation steps on a subsample of statements via a crowdsourcing experiment. This experiment included human annotators from the two cultures (U.S. and China), who were questioned in their native language (see Appendix C for details).

	Humans vs. Models
Step 1	0.45
Step 2 (P)	0.78
Step 2 (PS)	0.52

Table 2: Kappa agreements between aggregated annotators and aggregated models. Step 1 annotates the binary relevance of a statement for a given dimension. Step 2 indicates the magnitude of the dimension. Step 2 (P) uses three labels: $-1, 0, 1$; Step 2 (PS) uses five labels: $-2, -1, 0, +1, +2$.

The human annotated sample contains 40 statements per Hofstede dimension. From these, half (20) were considered by the LLMs as representative for the respective dimension, while half were labeled as not relevant. We also selected four control statements per dimension to measure the quality of the human labels.

The first annotation step, i.e., relevance to a given dimension, was conducted on a set of five people from U.S. and five people from China. We then selected the top three annotators per country, by evaluating the responses on the control statements and the needed time in seconds for each prediction. We aggregated these annotations using majority voting (i.e., a statement is considered relevant for a given dimension if at least 3/6 humans agreed).

The second annotation step, i.e., dimension polarity, considered only the statements marked as relevant for the given dimension in the previous step, i.e.: 24 samples for PDI, 26 for IDV, 21 for MAS, 30 for UAI, 19 for LTO and 21 for IND. Similar to the previous step, we selected top 3 annotators per country, and aggregated their responses using the median result.

Table 2 shows the agreement between the aggregated human and model labels for the two annotation steps. For the first step, the table shows moderate agreement (0.42) between the aggregated human and LLM labels. However, we observed that models are biased towards *higher precision* in their decisions: the LLMs had a precision of 79.63% relative to the human label. This indicates that the examples marked as relevant by the LLMs are considerably more likely to be correct than the moderate agreement indicates.

For the validation of the second annotation step, we considered two different settings: one in which we analyze only the polarity (P) (i.e., positive or

negative); and one where we also take into account the polarity strength (PS) (i.e., slight or strong). Table 2 indicates substantial agreement between humans and models in the P setting and moderate for the finer-grained PS one. For this step, we also measured mean absolute error (MAE) between the aggregated human labels and the LLM labels, on a -1 to $+1$ scoring scale for the polarity setting, and on a -2 to $+2$ scale for the strength-polarity setting. The reported MAE is 0.15 for the first setting, and 0.78 for the second setting. These results indicate that the final scores generated by LLMs are generally well aligned with the central tendency of culturally diverse human annotations.

5 Evaluation Methodology

Our overall goal in this work is to measure the cultural biases that LLMs may have in response to prompts that test a variety of societal aspects. To this end, we prompt each evaluated LLM in both English and Chinese on whether it agrees or disagrees with a given *CultureProbe* statement. The prompts used are listed in Appendix D. Based on the final responses, we compute an average score per dimension, which tells us the model's overall position relative to each Hofstede dimension.

6 Experimental results

6.1 Settings

We used four LLMs from four different families, two of which are developed in the United States, while the others are representative of China. The used models are GPT 5.4 (`gpt-5.4-2026-03-05`), Llama 3 (`llama-3-70b-instruct`), and DeepSeek-R1 (`deepseek-r1-0528`), and Qwen 2.5 (`qwen-2.5-72b-instruct`) respectively. Due to the variability in model responses throughout the entire labeling pipeline and evaluation, we selected the most frequent predictions from five different runs, using temperature 1.

6.2 Results and discussion

We present the scores of the four models when evaluated on the *CultureProbe* dataset. In our analysis, we also use the empirically measured Hofstede values of the two nations[5] (i.e., U.S. and China), derived from human measurements, as a comparison baseline. In the first two subsections we compare the models when prompted in their "native" languages (i.e., English for GPT and Llama, and Chinese for DeepSeek and Qwen). The third subsection presents the cultural differences that individual models exhibit across different languages.

6.2.1 Comparing U.S. vs. Chinese LLMs

Figure 1 shows a comparison between the aggregated responses from the Chinese models and the U.S. models, when prompted in their "native" languages. The aggregated response is the median of two LLMs per culture, where each LLM was run 5 times. We compare the aggregated LLM responses with the actual (human) Hofstede values of the two nations.[5] To highlight the sociocultural *differences* between the two LLM groups, we considered only the statements where the two groups responded differently for this experiment.[6] To align the scale of our experiments (-2 to 2) with the scale of the measurements for the human sociocultural differences (0 to 100), we then applied min-max scaling.[7]

As the figure shows, *culturally-distinct LLM groups follow trends similar to those observed among people from the same culture across 5/6 Hofstede dimensions*. For example, a higher PDI that is present for both Chinese human measurements and Chinese LLMs shows an acceptance for hierarchical structures and authority. In contrast, Western cultures with a lower PDI favor more egalitarian values. This is illustrated by the statement *"Unpaid internships should be banned"*, with which U.S. LLMs generally agree, while Chinese models tend to disagree. To detail, the U.S. LLMs consider unpaid internships as unequal treatment (thus, showing a low PDI), while the Chinese LLMs accept hierarchical differences and see unpaid internships as a privilege granted by those with power (i.e., higher PDI).

The most notable difference is observed for the IND metric. For example, the two LLM groups had contradictory predictions for the statement *"Developed countries should reduce the working year."*; the U.S. models agreed to the statement, while Chinese models rejected it. This difference highlights that the U.S. models reflect a sociocultural tendency toward indulgence (as defined by Hofstede),

[5] As measured by `https://www.theculturefactor.com`

[6] Recent research showed that LLMs tend to converge to similar outputs (Jiang et al., 2025). In this work, we focus on understanding the remaining differences. Inspecting *all* LLM outputs shows the same trends but attenuated by the large output similarity. See Appendix E for details.

[7] We have adopted this scalability solely for readability. We do not intend to directly compare the two measurements.

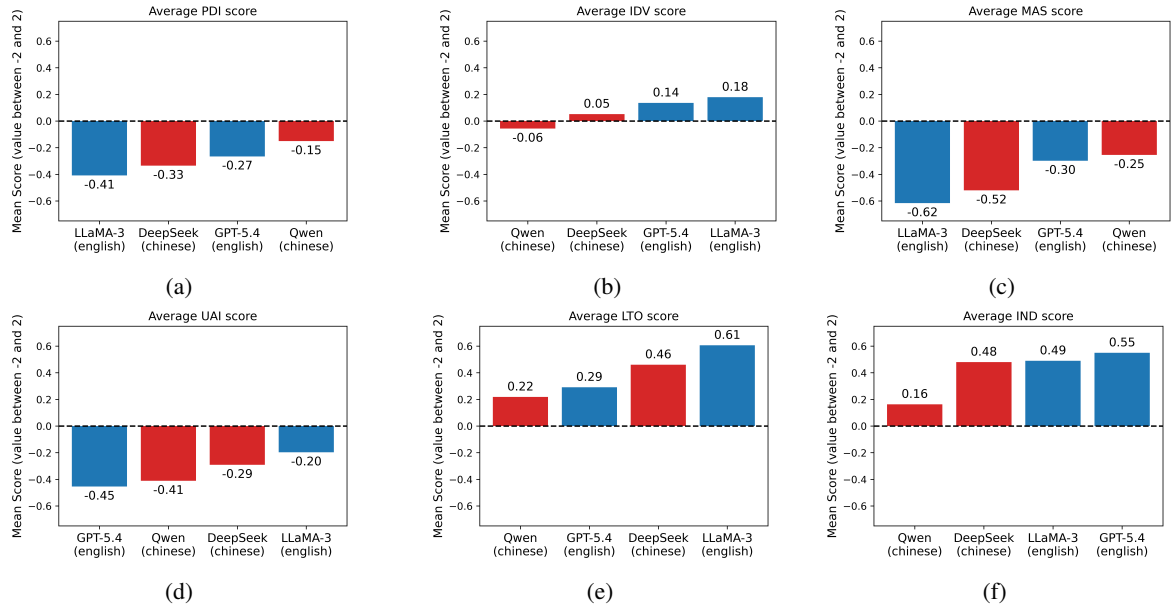


Figure 3: Average scores across all Hofstede dimensions for each model when prompted in its "native" language. The red bars represent Chinese models, while the blue bars represent U.S. models. Note that model outputs contain values in the interval $[-2, +2]$; however we used $[-0.75, +0.75]$ for the Y axis as it better fits the average scores.

whereas the Chinese models align more with values of restraint. See Appendix B for more examples.

Overall, this comparison indicates that the models do, in fact, reflect the sociocultural values of the countries in which they were developed. This is an important observation, given that these models are designed for international use, where many users come from cultures with different values and norms. This disconnect strongly suggests a pragmatic conclusion: a single builtin alignment model is not sufficient; we need LLMs that can accommodate multiple forms of alignment, each suited to the sociocultural context of their target users.

6.2.2 Comparing individual models on their "native" languages

Figure 3 compares the models used in our experiments when prompted in their respective "native" languages. In this experiment, we aggregate the scores from all statements relevant to each dimension, rather than focusing solely on the differences (as in the previous subsection).

We observe several deviations from the trends established by human measurements. For example, in Figure 3a, DeepSeek's PDI values are low and closely resemble those of the U.S. models. This contradicts the sociocultural values observed in contemporary China (Hofstede, 1991), where hierarchy is generally accepted and PDI scores are consistently high. Another outlier appears in Figure 3c, where DeepSeek shows a low MAS score, in contrast to the high score associated with Chinese culture, which reflects goal-oriented behavior and a clearer delineation of gender roles. DeepSeek also diverges from cultural expectations for IND, as shown in Figure 3f, where it exhibits a higher level of indulgence, close to the U.S. models. Overall, these results indicate a misalignment between DeepSeek's predictions and the Hofstede values associated with Chinese culture. In contrast, Qwen's predictions closely align with the sociocultural values of China across all dimensions.

There are multiple possible interpretations for the observed differences between Qwen and DeepSeek. For example, Jin (2023) presents the Chinese economic, social, and political model as complex and difficult to categorize on the spectrum between socialism and capitalism. The work characterizes Chinese culture as a heterogeneous hybrid system that combines socialism with Confucian principles (for the social fabric of the Chinese society), while also valuing traits typically associated with capitalist systems such as "fierce" competition and meritocracy (as the engine driving the country's economical progress). This shows long-term values with pressures for short-term optimizations and rapid adaptations, which are specific to western cultures. This may explain the misalignment seen in Figure 1 for the LTO dimension.

Jin (2023) suggests that Western sociocultural categorization systems (e.g., Hofstede dimensions) should be context-dependent (e.g., along authority,

family, market, or ideology axes). For example, Qwen's high PDI score may reflect Confucian values that remain prevalent in modern China in the context of of family or formal authority. In contrast, a low PDI score could indicate the influence of socialist ideology, which emphasizes equality. Similar observations can be made for the other dimensions as well.

Considering these observations, it is likely that the two Chinese models capture different, yet equally valid, perspectives of Chinese culture. A more mundane explanation is also possible: DeepSeek may have been developed with a more international user base in mind, where Western sociocultural values are perceived as dominant. Nevertheless, in aggregate, the two Chinese models closely reflect Chinese sociocultural values.

Similar patterns are observed for the U.S. models. Llama exhibits sociocultural values that are somewhat more left-leaning compared to GPT. However, when considered in aggregate, both models align well with U.S. sociocultural values.

6.2.3 Comparing individual models across different languages

Table 3 shows the average scores across all Hofstede dimensions for each model, when prompted in English and Chinese. The cells in green present cases where the cross-lingual sociocultural differences are statistically significant (p-values < 0.05 under a bootstrap[8] resampling significance test).

This experiment enables us to assess whether changing the prompt language influences the sociocultural values expressed by the model. We observe moderate cross-lingual differences within individual models, particularly for Qwen and GPT, where more statistically significant differences appear: for GPT in MAS and IND, and for Qwen in all dimensions except IDV. We consider this phenomenon normal, as it may indicate that the models are somewhat differently aligned for the two languages and the respective cultures.

In contrast, it is potentially concerning that for most other dimensions and models, no statistically significant differences are observed. This suggests that these models may culturally align their outputs to a common profile, regardless of differences in users' sociocultural backgrounds.

[8]We performed paired bootstrap resampling with 10,000 samples using a two-sided test.

Dim.	Lang.	GPT	Llama	DeepSeek	Qwen
PDI	English	-0.27	-0.41	-0.30	-0.05
	Chinese	-0.32	-0.40	-0.33	-0.15
IDV	English	0.14	0.18	0.07	-0.05
	Chinese	0.09	0.13	0.05	-0.06
MAS	English	-0.30	-0.62	-0.50	-0.13
	Chinese	-0.37	-0.47	-0.52	-0.25
UAI	English	-0.45	-0.20	-0.28	-0.54
	Chinese	-0.42	-0.25	-0.29	-0.41
LTO	English	0.29	0.61	0.46	-0.08
	Chinese	0.39	0.61	0.46	0.22
IND	English	0.55	0.49	0.41	0.30
	Chinese	0.42	0.48	0.48	0.16

Table 3: Average scores across all Hofstede dimensions for each model when prompted in both languages, evaluated on *CultureProbe* dataset. The scores represent inclinations towards the positive or negative polarity of the six dimensions. The green highlighting indicates statistically significant differences.

6.3 Potential explanations

Our analysis indicates that the four models studied exhibit cultural patterns directionally consistent with country-level differences in Hofstede's dimensions. This alignment may arise from two main sources: pre-training and instruction tuning/alignment. During pre-training, it is possible that model developers bias the web-crawled data toward their own language or sociocultural values. However, given the scale and diversity of the web, it is unlikely that pre-training datasets are deliberately manipulated in a substantial way.

A more plausible explanation is that sociocultural values are introduced during the instruction-tuning and alignment phase, which remains largely opaque. If this is the case, our findings underscore the importance of greater transparency in instruction tuning, including openness about the instruction datasets used and the training procedures.

7 Conclusions

We investigated how LLMs encode the sociocultural contexts of their human creators. We introduced *CultureProbe*, a dataset constructed from natural debate statements using a multi-step methodology. The dataset includes synthetic labels that quantify the Hofstede sociocultural dimensions reflected in LLM responses to these statements. We conducted a comparative multilingual analysis of two groups of LLMs developed in different countries (U.S. and China) and compared the results with human sociocultural values.

Our results indicate a directional correspondence of LLMs to the sociocultural values and norms of

the countries in which they were developed. Moreover, in general, LLMs do not exhibit statistically significant differences in sociocultural values when the prompt language is changed. These findings underscore the limited ability of current LLMs to adapt to the sociocultural backgrounds of their international users. This limitation should be a key consideration in the development of future LLMs, particularly during the alignment process. We anticipate that our study will contribute to the development of multiple culturally informed alignment strategies, each tailored to the cultural context of the target users.

Limitations

Our work quantifies and analyzes the sociocultural values expressed by LLMs developed in two culturally distinct regions (the United States and China), using the Hofstede dimensions as a reference framework. Our analysis is limited to two models per country when constructing aggregated predictions. This constraint is driven by the large number of model queries required for both dataset construction and evaluation and our limited academic budget.

Specifically, dataset annotation was performed using all four analyzed LLMs, each prompted in two languages (English and Chinese), with five runs per model–language combination. Starting from approximately 2,500 statements, this process resulted in roughly 500,000 model requests across dataset creation and analysis. Expanding the study to include additional cultural regions or a larger number of models per region would substantially increase the number of required requests and the overall cost.

We employ the same models from the four families in both the annotation and evaluation process. The benchmark may reflect these models' internal biases and sociocultural interpretations. Although manual annotation partially mitigates this concern, it covers only a subset of the dataset due to the high cost of human evaluation.

Ethics

This work analyzes the sociocultural values expressed by LLMs when responding to real-world debate statements collected from an online platform. Because these debates address complex societal issues, they may include sensitive or potentially offensive content, such as references to drugs, sexuality, religion, or war. We intentionally did not filter out such topics in order to preserve realism and to measure the sociocultural values expressed by LLMs as faithfully as possible. As the dataset is used solely for analytical purposes and the models are employed exclusively for inference, this work neither introduces new harmful content nor modifies the models in any way that could increase harm.

We use these cultures to identify sociocultural biases in LLMs. It is not our intention to compare cultures.

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A Dataset statistics and examples

We started with a set of 2,447 scraped debates, 1,380 of which remain after the automated vote-count filtering process. After the first annotation step, we identified 433 statements that are relevant for PDI, 425 for IDV, 467 for MAS, 315 for UAI, 165 for LTO, and 202 for IND. Table 4 offers examples of relevant statements for each dimension, associated with the assigned score. Even though this dataset may be considered small in the age of LLMs, it is considerably larger than existing work in this space (Cao et al., 2023; The Culture Factor, 2024).

We release the *CultureProbe* dataset publicly - https://github.com/vdorastie/LLMs-Cultural-Values. Due to licensing issues, we release statement ids (instead of the statement texts) together with scrapping software, which guarantees that our work is reproducible.

Statement	Dim.	Score
The Electoral College should be abolished.	PDI	-1
Organ donation should be mandatory.	IDV	-2
The primary focus of prisons should be rehabilitation and reintegration, not punishment.	MAS	-1
Health care providers should be mandated to get the flu vaccine.	UAI	1
Governments should push to implement 100% renewable energy.	LTO	2
Marijuana should be as legally accessible as alcohol.	IND	2

Table 4: Examples of statements that are relevant for each Hofstede dimension, with their assigned score. In case of agreement with the statement, a positive score shows alignment with the positive polarity of the listed dimension, while a negative score shows alignment with the negative polarity of the dimension. Note that a statement may be relevant for multiple dimensions.

We extract, via web scraping, a list of associated subjects for each statement in the dataset. As explained in section 4, the statements cover a broad area of domains. Figure 4 shows a histogram of 30 most frequent subjects in *CultureProbe* statements.

B Examples of sociocultural statements and LLM responses

In this section we analyze multiple statements, for different Hofstede dimensions, where the two LLM groups (U.S. and Chinese) responded differently.

A statement where the two groups responded differently is *"Every human should have the right and means to decide when and how to die"*. The U.S. LLMs' agreement reflects a more individualistic outlook, whereas the Chinese LLMs' disagreement aligns with the lower IDV score associated with Chinese culture, in which individual decisions are expected to consider the welfare of one's group and broader society.

For the statement *"The UK should remain in the EU if a hard Brexit is the only alternative option"*, the U.S. models' agreement aligns with a higher UAI, emphasizing stability and predictability, whereas the Chinese models' disagreement indicates a greater tolerance for uncertainty in pursuit of potential long-term benefits.

C Human annotation experiment

The human annotation experiment was conducted using Prolific (https://www.prolific.com/) for participant recruitment and compensation. The survey and annotation interface was created using Jatos (https://www.jatos.org/).

The experiment consisted of two stages. In the first stage, each annotator was required to determine whether a statement was relevant to a dimension, one dimension at a time. Therefore, the annotation process was simplified to a series of binary Yes/No questions. Before each new dimension, a brief explanation accompanied by examples was given to familiarize the annotator with the specific dimension. The second stage followed the same structure, but the annotators had 4 possible responses, similar to the LLM step 2 annotation process.

Participants were recruited directly through Prolific using Nationality and Place of Birth screening criteria (i.e., from U.S. and China). Each participant was questioned in its native language. All annotations were collected anonymously. Participants received £9.00/hr., with an expected completion time of 1 hour and 20 minutes for the first task, and 40 minutes for the second task. We recruited five people from China and five people from U.S. for the first stage, and five people from U.S. and four people from China for the second stage. In both stages, we used in our analysis only the top three annotators from each culture, who were chosen based on their performance on the control statements. Table 5 presents the inter-annotator average Kappa agreement for the two steps.

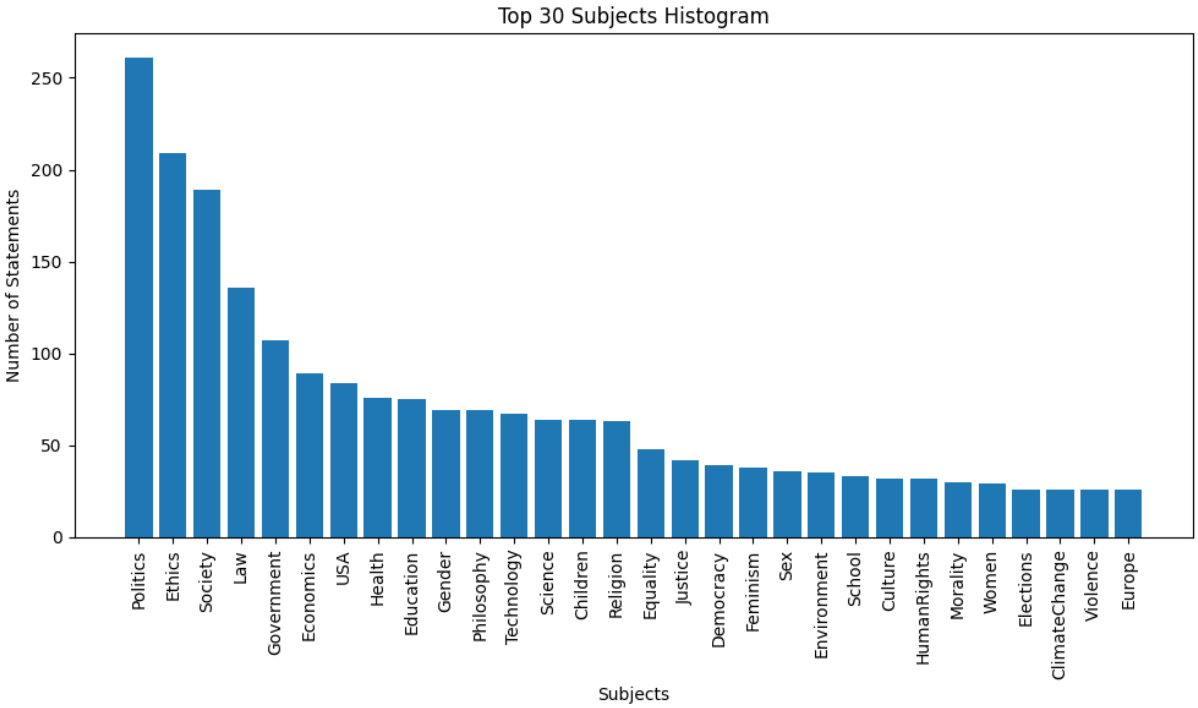


Figure 4: Top 30 most frequent subjects in *CultureProbe*. Each statement may have one or multiple associated subjects. We observe that the dataset is formed of general statements that cover a wide area of subjects of interest.

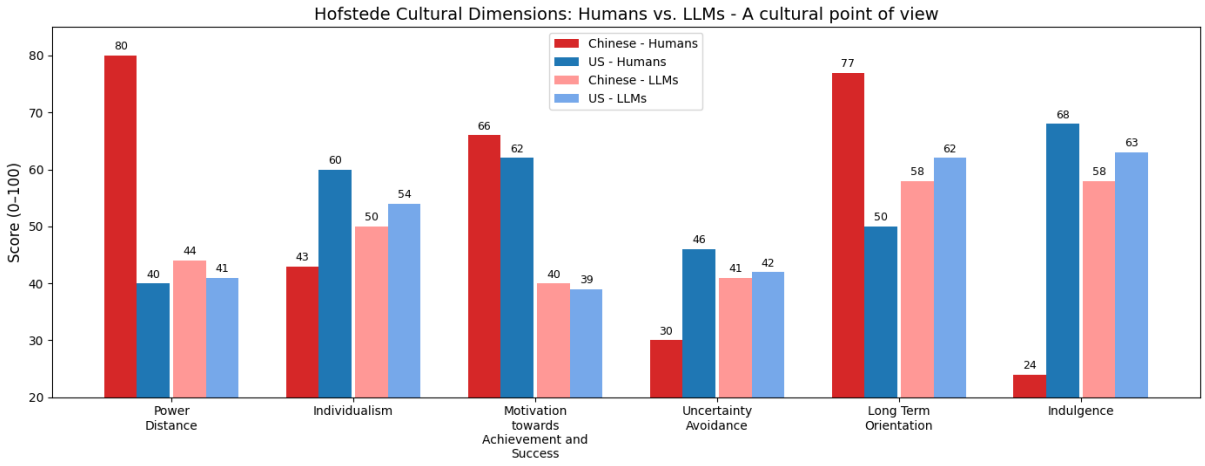


Figure 5: Scoring of humans and LLMs according to the Hofstede dimensions, considering all LLM outputs. LLMs were prompted to agree or disagree with a given statement, and each response was assigned a polarity score ranging from -2 to $+2$ along the corresponding Hofstede dimension. These scores were then averaged and linearly mapped to a 0–100 scale to ensure comparability with human measurements. We observe that the LLMs largely follow the trends observed in human measurements across five out of six dimensions.

D Prompts used for LLM evaluation

Figures 6 and 7 show the prompts used during the evaluation methodology, where we ask LLMs whether they agree or disagree with a given *CultureProbe* statement.

```
You are given a claim. Your aim is to reason on that claim, by
constructing arguments for and against it. You are then required to
respond with a single word from the list: [true, false], agreeing or
disagreeing the given claim.

You have to output only the final answer, not the whole reasoning
process. Respond only in this JSON format:
{"answer": true}

Below you are presented the claim:
```

Figure 6: English prompt used during the evaluation experiment.

E U.S. vs Chinese LLMs on all datapoints

Figure 5 presents the differences between the humans of the two analysed cultures and the LLMs, considering all LLM outputs, not only the differ-

Step	Inter-annotator
Step 1	0.25
Step 2 (P)	0.59
Step 2 (PS)	0.33

Table 5: Average Kappa agreement between annotators for all dimensions. Step 1 annotates the binary relevance of a statement for a given dimension. Step 2 indicates the magnitude of the dimension. Step 2 (P) uses three labels: $-1, 0, 1$; Step 2 (PS) uses five labels: $-2, -1, 0, +1, +2$.

```
你将会得到一个论断。你的任务是对该论断进行推理，构建支持和反对的论点。然后，
你需要从列表中选择一个词作为回应：\[true, false]，表示你是否同意该论断。

你只能输出最终的答案，不要输出完整的推理过程。仅以以下 JSON 格式作答：
{"answer": true}

下面是你要评估的论断：
...
```

Figure 7: Chinese prompt used during the evaluation experiment.

ent predictions of the LLMs. We observe that the LLMs still follow the human trends for 5/6 Hofstede dimensions, but the differences are attenuated.